

## API Abuse Risk Management

### Definition

API Abuse is the misuse of an application's APIs by sending excessive, automated, or malicious requests. This can lead to spam, unauthorized access attempts, data theft, service disruption, or increased server costs.

## Examples in the Internship & Job Portal

### Spam Job Applications

A bot submits thousands of job applications in a short time, creating fake applications and wasting recruiters' time.

### Brute-Force Login Attacks

Attackers repeatedly try different passwords on the login API to gain unauthorized access.

### Data Scraping

Automated programs repeatedly access job, company, or student APIs to copy platform data.

### AI Feature Misuse

Users send excessive requests to AI resume analysis or recommendation APIs, increasing server load and cost.

### Denial of Service

Large numbers of API requests slow down or temporarily make the platform unavailable.

### Impact

- Server performance degradation
- Fake job applications and spam
- Unauthorized access attempts
- Data theft and privacy violations
- Increased infrastructure and AI service costs
- Poor user experience

### Mitigation / Resolution

| Security Measure   | Purpose                                                                       |
|--------------------|-------------------------------------------------------------------------------|
| Rate Limiting      | Restrict the number of requests a user or IP can make within a specific time. |
| JWT Authentication | Allow only authenticated users to access protected APIs.                      |

|                                  |                                                                 |
|----------------------------------|-----------------------------------------------------------------|
| Role-Based Access Control (RBAC) | Ensure users can access only the APIs permitted for their role. |
| CAPTCHA                          | Prevent automated bots from submitting requests.                |
| Input Validation                 | Reject invalid or malicious API requests.                       |
| IP Blocking                      | Block suspicious IP addresses showing abusive behavior.         |
| Monitoring & Audit Logs          | Track API usage and detect unusual activity.                    |
| HTTPS Encryption                 | Protect data transmitted between clients and the server.        |

## Conclusion

API Abuse is a significant security risk for an internship and job portal because it can affect login, job applications, resume uploads, AI services, and company data. Implementing authentication, rate limiting, RBAC, CAPTCHA, monitoring, and encryption helps protect the platform and ensures reliable service for students, colleges, companies, and administrators.